Warhammer 40K Tournament Analysis

Does Army Cost Impact Competitive Performance?

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Abstract

This study investigates the relationship between financial investment and competitive performance in the Warhammer 40,000 World Championship 2025 by analyzing data from 402 players and their army compositions. By integrating tournament results with official miniature pricing data, the research evaluates whether army cost significantly impacts success and examines how strategic factors such as faction choice, detachment selection, and unit composition influence outcomes.
Results reveal a weak but statistically significant positive correlation between army cost and performance, suggesting that higher spending provides only a marginal advantage. However, no significant cost differences are found between top-ranked and bottom-ranked players, indicating that reaching elite competitive levels does not require greater financial investment. Furthermore, faction popularity shows no relationship with win rates, while detachment choice emerges as a key strategic factor. Unit-level and army archetype analyses (elite vs. horde) indicate overall balance within the meta, with limited faction-specific exceptions. Overall, the findings suggest that Warhammer 40,000 is not a pay-to-win game: although economic factors affect accessibility and provide slight advantages, competitive success is primarily driven by player skill, strategic decision-making, and preparation.

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Introduction

This project analyzes data from the Warhammer 40,000 World Championship Tournament 2025 held in Atlanta, Georgia, to investigate the relationship between financial investment and competitive performance in tabletop wargaming. By examining 402 tournament participants and their army compositions, we seek to understand whether monetary costs influence success in high-level competitive play and how faction dynamics shape the competitive meta.

1.1 What is Warhammer 40k?

Warhammer 40,000 is a tabletop miniature wargame produced by Games Workshop where players command armies of intricately detailed plastic miniatures in tactical sci-fi battles. The hobby combines three core elements: collecting miniatures, assembling and painting them, and playing strategic games on terrain-filled tabletops. Players construct army lists by selecting units from their chosen faction's roster, with each unit assigned a point value reflecting its battlefield effectiveness. Tournament games typically use a 2000 point limit, requiring players to balance troop choices, elite units, vehicles, and hero characters within this budget while adhering to army composition rules. The relationship between competitive play and hobby costs is significant. A tournament ready 2000 point army requires substantial monetary investment: players must purchase multiple unit boxes, hero models, and often vehicles or large centerpiece models. Individual infantry units can cost 30-60 euros, while tanks and elite units range from 50-150 euros. A complete competitive army typically represents an investment of 500-1500 euros before considering paints, tools, and hobby supplies. Tournament success often correlates with access to newer, competitively priced units, creating an economic dynamic where competitive players may need to regularly update their collections as the game's meta evolves through new releases and rule updates. This intersection of hobby investment and competitive viability makes the tournament scene both a test of tactical skill and a reflection of the financial barriers inherent to the miniature wargaming hobby.

1.2 Research Question

This study investigates multiple dimensions of competitive Warhammer 40,000 to understand what factors drive tournament success. Our analysis addresses four interconnected research questions:

- **Economic Impact on Performance**
  Does the monetary cost of an army influence competitive outcomes? By analyzing army list costs at both individual composition and faction levels, we examine whether players who invest more in miniatures achieve better tournament results. This addresses a fundamental question for the competitive community: is tournament success influenced by economic barriers to entry, or do skill and strategy remain the primary determinants regardless of collection costs?

- **Meta Analysis: Popularity vs. Performance**
  Does playing a popular faction provide a competitive advantage over niche choices? We analyze whether the most commonly selected factions in the tournament achieve better win rates than less popular alternatives, examining if meta popularity correlates with competitive viability. Additionally, we investigate whether specific detachment choices within each faction offer strategic advantages, determining if certain subfaction builds are statistically superior to others.

- **Unit Analysis: Elite vs. Horde Army Composition**
  Is it more advantageous to field many low-cost units (horde armies) or fewer high-cost units (elite armies)? By analyzing the relationship between unit count, point distribution, and tournament performance, we determine whether specific army composition strategiesconcentrating points in powerful units versus spreading them across numerous cheaper unitsoffer competitive advantages.

- **Faction Dynamics and Matchup Interactions**
  Does the game exhibit "rock-paper-scissors" dynamics where specific factions hold inherent advantages over others? By examining head-to-head matchup data between popular factions, we seek to identify hard counters and assess whether competitive success depends on favorable faction pairings or remains primarily skill-based across all matchups.

Datasets

To analyse this topic we collected and integrated data from two data sources, obtaining two dataset to work with.

2.1 Dataset 1: Tournament Data

The first dataset was obtained from a web page that records datas from all relevant Warhammer torunaments (`https://listengine-v3-be.onrender.com`). The extraction was done via REST API and a custom parsing pipeline was created to store them in a convenient format. In particular the data we selected are from the 402 players that took part in the Warhammer 40,000 World Championships 2025 (Atlanta, GA, November 2025).

Format: JSON responses from API, processed through multiple parsing stages.

Content:

- Event information (name, date, location)
- Player information (names, IDs, rankings)
- Faction and detachment choices
- Match history with opponents and results
- Complete army lists unit compositions with:
  - Categorization (Characters, Battleline, Dedicated Transports, Other Datasheets)
  - Unit names and point costs
  - Enhancements and warlord designations

When interpreting these data, it is important to keep two aspects in mind: how victory is determined in a Warhammer 40,000 match and how tournament rankings are calculated. A match winner is decided by the total number of Victory Points scored over five game turns. Victory Points can be earned in two main ways: by controlling primary objectives on the battlefield (five objectives are placed across different zones, each granting points every turn when controlled for a maximum of 50 points) and by completing secondary missions drawn at the start of each players turn, such as destroying an enemy unit or holding the central objective, for a maximum of 40 points; an additional 10 points are awarded for having a fully painted army. Furthermore, the ranking of the World Championships 2025 is not based solely on the number of wins but also considers the Path to Victory, which evaluates the strength of the opponents faced: defeating strong opponents who achieved good results leads to a higher ranking, while beating weaker opponents who performed poorly results in a lower ranking even with the same number of wins.

2.2 Dataset 2: Pricing Data

The second dataset was obtained through direct contact with Arsmanufactorea, an official Warhammer miniature retailer that shared the official price list and gave us permission to use them for research purposes.

Format: XLSX (Excel spreadsheet).

Content:

- Official unit prices in EUR
- Product codes and standardized names
- Unit type classifications
- Faction name

These data originate from a product catalog used to organize an online marketplace, therefore the Faction Name category does not necessarily correspond to the classifications used in tournaments. In fact, several units can be fielded in multiple armies, and some detachments allow players to include reinforcement units from entirely different factions. This distinction is taken into account in the subsequent integration procedures.

Data Acquisition Process

The data acquisition phase is foundamental for this research as no single repository contains both competitive performance metrics and economic data for Warhammer 40K armies, since both points and prices continue to change.

3.1 Tournament Data Extraction

Tournament data were obtained from the online repository for competitive Warhammer 40K results by leveraging ListEngines REST API, which provides hierarchical access to event, player, and list-level information. We developed a custom Python extraction pipeline structured into four sequential phases: event discovery, player-level retrieval, data parsing and normalization, and Data Validation.

Phase 1: Event-Level Data Acquisition
Data collection began with a single call to the event endpoint, returning comprehensive metadata including event name, date, location, total number of players, and the complete roster of participant identifiers. This response formed the foundation of the extraction workflow, as player IDs were required for subsequent list retrieval. Event data were retrieved via a standard GET request implemented with Pythons requests library, including basic timeout handling and connectivity safeguards.

Phase 2: Iterative Player List Collection
The core of the extraction pipeline consisted of retrieving detailed army lists and match data for all 402 participants. Each player required a dedicated API request to a player-specific endpoint. To respect implicit API rate limits and avoid server overload, we implemented an iterative loop with a conservative 0.5-second delay between calls.

This phase presented several operational challenges. Potential issues included request failures, server timeouts, temporary unavailability, and malformed responses. To address these, each request was given a 10-second timeout, and failures triggered structured exception handling that logged the error while allowing the pipeline to proceed without interruption. A real-time progress tracker displayed player index, name, and request status, enabling early detection of anomalies during execution.

Phase 3: Data Parsing and Normalization

The raw JSON responses featured deeply nested structures, requiring careful transformation into analysis-ready format. Player-level metadata (rank, name, faction, detachment, tournament score) was first extracted. Army lists were then parsed while preserving their categorical structure (Characters, Battleline, Dedicated Transports, Other Datasheets), and unit-level attributes such as points, enhancements, and warlord designation were captured.

Unit names occasionally included special characters or inconsistent formatting and faction names appeared in multiple capitalization variants (e.g., space marines, Space Marines), necessitating systematic normalization. These issues were progressively addressed to ensure resilience and consistency in the final dataset.

Phase 4: Data Validation

To ensure extraction accuracy, we implemented several validation checkpoints throughout the pipeline. After retrieving event-level data, we verified that the reported player count matched the length of the returned player roster. During individual list retrieval, we tracked success rates and flagged any players whose lists failed to download, allowing us to attempt manual recovery if necessary. Following parsing, we performed structural validation to confirm that all expected fields were present and properly typed, for example, verifying that point costs were numeric values rather than strings, and that categorical assignments matched expected values. All raw JSON responses were saved to disk with timestamps and player identifiers, creating a permanent record of the source data. This preservation served multiple purposes: it enabled validation of our parsing logic by comparing processed data against original responses, provided a backup in case of processing errors, and documented the exact state of the tournament data at the time of collection, which is important given that online platforms can occasionally modify or remove historical data.

3.2 Technical Implementation Details

Our extraction script was implemented as a modular Python application with clearly separated concerns. The scraping_warhammer() function handled event-level requests, get_player_list() managed individual player queries, and scraping_complete_with_lists() orchestrated the overall extraction workflow. We used the requests library for all HTTP communications, implementing appropriate headers and timeout configurations. The json module handled serialization and deserialization of API responses, while helper functions managed data normalization tasks such as standardizing army faction names. Error handling was implemented at multiple levels: network-level exceptions caught connectivity issues, HTTP status code checking identified API errors, and JSON parsing exceptions handled malformed responses. Each type of error was logged with sufficient detail to enable diagnosis and potential recovery. The script also implemented a graceful degradation strategyif a single player's list failed to download, the extraction continued for remaining players rather than terminating entirely. The entire extraction process required approximately 1 hours to complete, primarily due to the rate-limiting delays between API calls. The resulting dataset comprised 402 complete player records with detailed army compositions, representing a comprehensive snapshot of the highest level of competitive Warhammer 40K play at the end of 2025.

3.3 Pricing Data Import

The pricing dimension of our research required accessing proprietary commercial data, as Games Workshop does not publish comprehensive programmatic price lists. We established contact with Arsmanufactorea, an authorized Warhammer retailer in Italy, who agreed to share their official price database for academic research purposes. The retailer provided their complete product catalog in XLSX format, which was a very large dataset containing numerous fields, many of which were unnecessary for our analysis (e.g., product packaging dimensions, warehouse codes, and other administrative attributes). To simplify future analyses and keep the dataset focused, we retained only the relevant columns: catalog faction, standardized product name, unit points, and price in euro. This dataset represented current market prices as of November 2025, contemporaneous with the tournament data.

Data Integration and Cleaning

The data integration and cleaning phase represents the core of the project, as it enables the transformation of heterogeneous raw sources into a coherent, analysis-ready dataset. The two original datasets, tournament data extracted via API and pricing data provided by a commercial retailer, presented multiple forms of heterogeneity at both schema and instance level, requiring systematic preprocessing and integration strategies.

4.1 Unit Name Standardization

One of the main integration challenges was the alignment of unit names between tournament army lists and the pricing dataset. Tournament data relied on player-defined naming conventions, often including abbreviations, informal terminology, variable capitalization, and additional descriptors related to equipment or enhancements. Conversely, the pricing dataset followed standardized commercial naming conventions.

To address this name heterogeneity, a multi-step normalization process was applied. Unit names were converted to lowercase, leading and trailing whitespace was removed, and multiple internal spaces were collapsed into a single space. Special characters and formatting inconsistencies were systematically eliminated.

This normalization enabled deterministic matching in the majority of cases.

For residual mismatches, a curated mapping dictionary was introduced. This dictionary encoded expert knowledge about equivalences between tournament unit names and retail product names and was reused across executions to ensure consistency and reproducibility. This approach was preferred over purely distance-based fuzzy matching techniques, as many discrepancies were semantic rather than typographical and could not be reliably resolved using similarity scores alone.

4.2 Price Assignment and Unit Size Logic

After resolving naming conflicts, the integration process addressed a second major source of heterogeneity: differences in unit size and point values. Tournament army lists often include units with point values exceeding those of the base retail product due to increased squad size or optional enhancements.

The pricing dataset provides prices for base unit configurations only. To bridge this gap, we designed a rule-based price-scaling function that estimates unit cost based on the ratio between tournament point values and base point values in the pricing dataset.

The logic follows these principles:

- when the tournament unit point value matches the base reference, the base retail price is applied;
- when the point increase is within +40 points, the difference is attributed to enhancements and the base price is retained;
- when the point value exceeds this threshold, the price is scaled proportionally to reflect the need for additional models;
- the final unit cost is computed by multiplying the estimated single-unit price by the quantity of units included in the army list.

This approach provides a realistic economic approximation of army compositions while maintaining consistency across factions and unit types.

4.3 Deduplication and Match Integration

Tournament match data presented instance-level duplication, as each match was recorded once per participating player. To resolve this issue, matches were reconstructed by merging player-centric match records and retaining a single representation per match.

This deduplication step eliminated redundant records and ensured that match outcomes were represented consistently. The resulting match table preserves all relevant information while avoiding statistical distortions in downstream analyses.

4.4 Data Quality Assessment and Improvement

In line with the Data Management pipeline discussed during the course, data quality was evaluated both before and after the cleaning and integration process. The assessment focused on the quality dimensions most relevant to the projects objectives and data characteristics: completeness, consistency, redundancy, and syntactic accuracy.

- Completeness Completeness was measured as the proportion of non-missing values in key analytical attributes, with particular attention to unit-level economic information.

In the raw tournament dataset, economic attributes were entirely absent, resulting in 0% completeness for unit prices. After integration with the pricing reference dataset and the application of rule-based price estimation, price values were successfully assigned to all army list units included in the final database.

As a result, the completeness of unit-level price information increased to 100%. It is important to note that this completeness refers to the presence of values, not their absolute accuracy, as some prices are estimated rather than directly observed.

- Consistency Consistency was evaluated at both instance level and structural level.

At the instance level, tournament match data exhibited systematic duplication, as each match was recorded once per participating player. Matches were therefore reconstructed by merging player-centric records and retaining a single representation per match. For tied matches, a deterministic rule based on player identifiers was applied to avoid symmetric duplicates. After this deduplication step, each match was represented exactly once in the final dataset.

At the structural level, referential consistency was enforced through relational constraints in the database schema. All foreign key relationships linking tournaments, players, matches, army lists, and army list units were validated, achieving 100% referential integrity in the final database.

- Syntactic Accuracy Syntactic accuracy was assessed by measuring the proportion of unit names that could be successfully matched between the tournament dataset and the pricing reference dataset.

Before normalization, approximately 69% of unit names could be matched automatically. After applying systematic normalization procedures and introducing a curated mapping dictionary encoding domain-specific equivalences, the matching rate increased to 100%. This eliminated residual ambiguities and ensured reliable integration across all units.

- Redundancy Redundancy was assessed on the pricing reference dataset by measuring the proportion of duplicated records.

Before cleaning, the pricing dataset exhibited a non-negligible level of redundancy. These duplicated entries did not represent data errors, but rather reflected the fact that the same unit can legitimately appear in multiple army factions. While this duplication is acceptable and even necessary at the schema level, it introduces redundancy when performing analytical joins.

To address this issue, a deduplicated version of the pricing dataset was created by retaining a single reference entry per unit name and base configuration. After preprocessing, the redundancy rate from 39.68% was reduced to 0%, ensuring minimality and preventing artificial inflation of join results during analysis.

Quality Improvement Summary Overall, the data preparation pipeline led to clear and measurable improvements in data quality. Completeness increased through systematic enrichment, consistency was enforced via deduplication and relational constraints, redundancy was reduced to ensure minimal analytical representations, and syntactic accuracy reached full coverage through controlled normalization strategies.

These improvements ensured that the final dataset was fit for analysis with respect to the projects research questions.

Data Storage

After the data integration and cleaning phase, the final dataset was organized and stored in a relational SQLite database. Rather than maintaining a single denormalized table, the data were decomposed into six logically distinct tables (tournaments, players, matches, army_lists, prices and army_list_units) each representing a specific entity of the tournament domain.

A relational database model was selected for data storage, as the integrated dataset exhibits a highly structured schema with well-defined entities and stable relationships. Tournaments, players, matches, and army compositions are naturally represented through relations connected by primary and foreign keys, making the relational paradigm particularly suitable.

Alternative solutions such as document-oriented databases (e.g., MongoDB) or graph databases (e.g., Neo4j) were considered but deemed less appropriate for this context. The data do not present significant schema variability over time, nor do they require nested or unstructured representations that would justify a document-based model. Similarly, while relationships between entities exist, they are mostly hierarchical and transactional rather than highly interconnected or exploratory, reducing the benefits of a graph-based approach.

The relational model provides strong guarantees in terms of data integrity, explicit constraint enforcement, and clarity of representation, which are essential for reproducible analysis and data quality assessment. Moreover, the relational structure facilitates efficient joins and aggregations, which align well with the analytical objectives of the project.

This relational decomposition was designed to reduce redundancy, improve clarity, and explicitly encode the relationships between tournaments, players, matches, and army compositions. Primary keys uniquely identify each entity, while foreign keys define the logical connections across tables, enforcing referential integrity and preventing inconsistent records.

The resulting database schema reflects the conceptual structure of the domain and supports transparent data validation and documentation. A detailed overview of the database schema, including attributes, primary keys, and foreign key relationships, is provided in Appendix (*Table* 19).

The relational database was generated through a Python-based pipeline that consolidates, validates, and structures the integrated data into a reusable format. Indexes were created on selected attributes frequently used for filtering or joiningsuch as tournament identifiers, player identifiers, army types, and unit namesto support efficient data access and ensure scalability in case of database-level querying.

The resulting database represents a reusable data layer, from which tables can be extracted and analyzed according to different analytical needs.

Exploratory Data Analysis

aA fundamental preliminary phase is Exploratory Data Analysis (EDA). Its main goal is to understand the main characteristics of the dataset, identify patterns, and help formulate the first hypotheses to be verified in another phase. The analysis then focused on:

- Faction popularity
- Player win rates
- Army Costs
- Price-to-point ratio per unit
- Score differentials

6.1 Faction popularity

Army	Count	%
Space Marines	32	7.96
Aeldari	31	7.71
Adeptus Mechanicus	28	6.97
Death Guard	23	5.72
World Eaters	23	5.72
Imperial Knights	20	4.98
Necrons	18	4.48
T'au Empire	16	3.98
Chaos Space Marines	16	3.98
Adepta Sororitas	16	3.98
Orks	14	3.48
Chaos Knights	13	3.23
...others (15+ factions)	—	—

Table 1: Faction Distribution

The analysis of faction popularity reveals a moderately skewed distribution. Space Marines (7.96%) and Aeldari (7.71%) lead the field with 32 players and 31 players, while the mean player count per faction is 14.89, which is slightly higher than the median of 13.0.

This difference between the mean and the median suggests that a small number of highly popular factions (the top 3-4) are pulling the average up, while the majority of factions cluster around a lower usage rate (between 2.5% and 4%). The tail of the distribution includes choices like Deathwatch and Thousand Sons, which have very limited play (<2%).

6.2 Player win rate

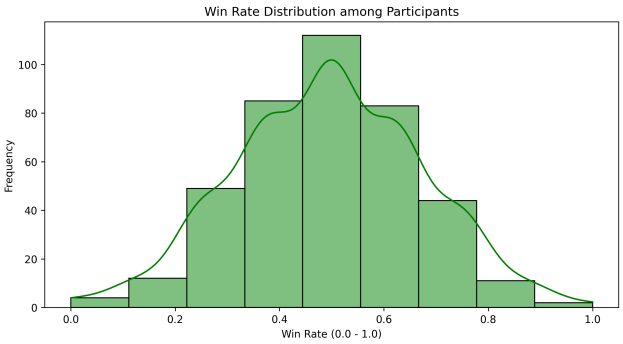


Figure 1: Win Rate Distribution among Participants

The analysis of player win rates indicates a highly balanced competitive environment. The distribution is unimodal and symmetric, peaking at the 50% mark.

The fact that the mean (0.49) and median (0.50) are essentially the same suggests that there are no significant outliers distorting the competition. Most players won roughly half their games, while very few players went undefeated or lost every match. This suggests that the skill gap between the bulk of participants is relatively narrow.

6.3 Army cost

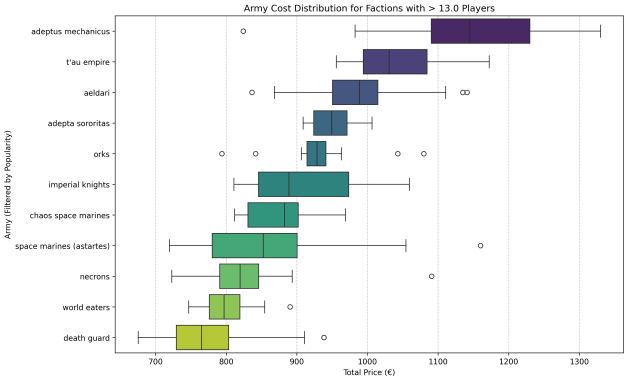


Figure 6.2: Army cost distribution for factions

The boxplot illustrates the cost distribution for the most popular armies. The graphic was filtered to include only factions with more than 13 participants (the median player count) to avoid skewed data from low-sample sizes.

Within this subset, we observe a massive economic disparity. There is roughly a 400 gap between the median cost of the cheapest popular faction (Death Guard) and the most expensive (Adeptus Mechanicus).

In conclusion, it can be said that certain factions tend to be significantly more expensive than others.

6.4 Price-to-Point per unit

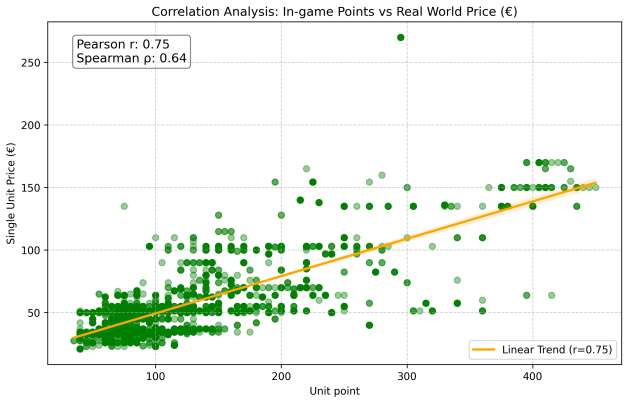


Figure 6.3: Correlation analysis points vs price

The analysis of the price-to-point ratio is a crucial aspect of Warhammer. Since players are strictly limited to 2000 points per army, the point value of each unit rules the army's composition.

The strong positive correlation (Pearson r = 0.75) confirms the trend that 'Points cost Money': units with high in-game points generally cost a higher price. This suggests that elite, powerful units are naturally more expensive.

This correlation reflects a fundamental game balance mechanism: high point costs limit the number of powerful units allowed in an army. Consequently, players face a strategic choice: fielding a compact 'Elite' army (few troops with high point values), a 'Horde' army (many units with low point values), or a balanced mix.

6.5 Score differentials

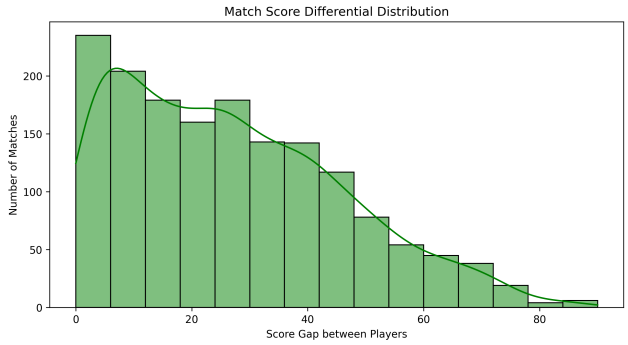


Figure 6.4: Match score distribution

The distribution of the score differential serves as an indicator of the competitive balance of the tournament.

The graph shows a healthy competitive environment. The majority of matches, which are characterized by a low score differential (On the left side of the graph), suggest that players were generally matched against opponents of similar skill levels.

Economic analysis

7.1 The current meta is pay-to-win?

Assuming that the most expensive units tend to be the strongest, a fundamental question emerges: Do the most expensive armies translate into superior competitive performance? The objective of this first subsection is to investigate if a higher financial investment correlates with better final rankings and win percentages.

Statistic	Value (€)
Mean	927.01
Std. Deviation	127.07
Minimum	581.00
25% (Q1)	832.75
50% (Median)	922.75
75% (Q3)	999.81
Maximum	1,330.00

Table 2: Descriptive Statistics of Army Costs

Firstly, an analysis of the 402 participating armies shows an average investment of around 925.69€. Although the absolute range is from a minimum of 581.00€ to a maximum of 1,330.00€, the standard deviation of 127.07€ indicates that the distribution is actually relatively concentrated. This suggests that, although outliers exist, the financial barrier to entry for most competitive players is closely around the mean.

To determine whether spending differences, even within this concentrated range, affect outcomes, the correlation between total army cost and performance metrics (rank and wins) was examined.

Cost vs Rank Position	Result
Pearson Correlation (r)	-0.1600
Pearson p-value	0.0013
Spearman Correlation (ρ)	-0.1393
Spearman p-value	0.0052

Table 3: Correlation Analysis: army cost vs rank position

Cost vs. Number of Wins	Result
Pearson Correlation (r)	0.1308
Pearson p-value	0.0086
Spearman Correlation (ρ)	0.1075
Spearman p-value	0.0312

Table 4: Correlation Analysis: army cost vs number of wins

The statistical tests reveal a clear trend:

- **Cost vs Rank**: A Pearson correlation of ρ = -0.16 (p = 0.0013). The negative sign indicates that as the cost increases, the rank number decreases.
- **Cost vs Wins**: A Pearson correlation of ρ = +0.131 (p = 0.0086). The positive sign confirms that higher costs are associated with a greater number of victories.

To provide a clearer perspective on the situation, the following two charts are presented below.

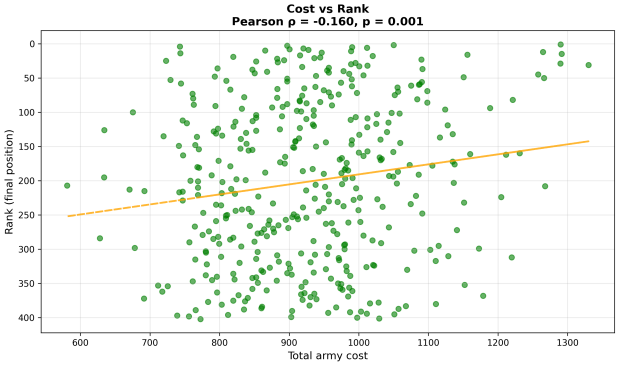


Figure 7.5: Cost vs rank

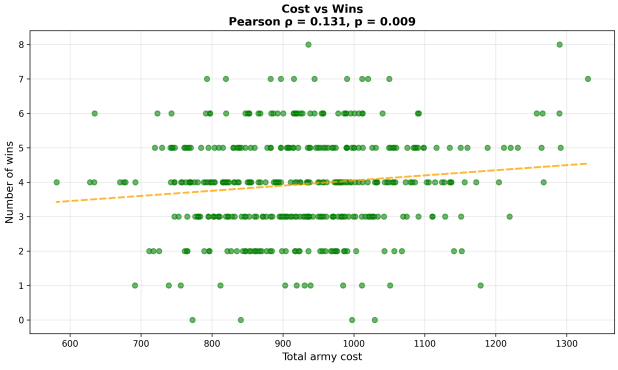


Figure 7.6: Cost vs wins

Both tests yielded a statistically significant p-value ($p < 0.05$), indicating that the relationship is not due to random chance, but there is a measurable link between money spent and competitive success. However, the correlation coefficients are relatively low with a $|\rho| < 0.2$, indicating a weak correlation.

While the data confirm that financial investment provides a competitive edge, it is not the only factor that determines victory. It's important to acknowledge that this analysis is limited to a dataset of 402 players from a single tournament. While this represents a significant sample for a single event, a wider dataset might have revealed different patterns. In conclusion, while financial investment in army units plays a small part in overall performance, player skill and strategy remain the dominant factors in determining the outcome.

7.2 Is a significant financial investment a prerequisite for reaching the top tier of competitive play?

This second question regarding economic analysis aims to determine whether investment is a necessary prerequisite for entering the top 25. Specifically, do the players with the best performance (Top 25) spend significantly more than those with the worst performance (Bottom 25)?

To determine whether reaching the highest competitive level requires a significantly higher budget, we isolated the two extremes of the ranking. Then, the costs of the top 25 players' armies (the tournament elite) were compared with those of the bottom 25 players (those with the lowest rankings). Specifically, the following results were obtained:

Top 25 Players (Rank 1–25)	Value (€)
Average Cost	968.43
Median Cost	936.00
Standard Deviation	155.94
Minimum Cost	723.00
Maximum Cost	1,291.50

Table 5: Descriptive Statistics: Top 25 Players

Bottom 25 Players (Rank 377–402)	Result (€)
Average Cost	928.91
Median Cost	955.85
Standard Deviation	111.79
Minimum Cost	739.00
Maximum Cost	1,110.75

Table 6: Descriptive Statistics: Bottom 25 Players

The statistical tests reveal:

- **Top 25**: The average army cost for the best players is 968.43 €. However, the median cost, which filters out the outliers, is about 936.00 €.
- **Bottom 25**: The average cost for the lowest ranked layer is 928.91 € with a higher median of 955.85 €.

At first glance, while the Top 25 average is roughly higher (40 €), the median actually reveals an inverted trend, where the elite players spent 20 € less than the typical player at the bottom of the ranking.

To verify whether the observed difference in average spending is real or merely coincidental, an independent T-Test was conducted. The result of the test yielded a p-value of $0.3082 > 0.05$. This means the difference in spending is statistically non-significant.

This analysis provides crucial insight into the tournament's economy. The data proves that there is no statistical evidence that spending more money is necessary to enter the Top 25. A player at the top of the leaderboard is just as likely to have spent the same amount (or even less, considering the median) as a player at the bottom. This suggests that the "barrier to entry" for the elite tier is defined by skill and game knowledge rather than a higher financial investment.

7.3 Summary of economic findings

Crossing the results obtained in the global correlation data (analysis 8.1) and the specific top-tier comparison (analysis 8.2), a definitive answer can be formulated regarding the role of the budget in competitive Warhammer 40k.

The data suggest a dual reality:

- **The general trend**: Considering the general trend, money offers a small statistical advantage. The weak positive correlation across the full dataset indicates that accessing expensive units can provide a marginal boost to performance, potentially helping an average player secure a few more wins.
- **The elite reality**: However, financial investment is not the determinant factor for excellence. The direct comparison between the Top 25 and Bottom 25 proves that reaching the podium does not require outspending. The best players in the tournament spent almost the same as the lowest ranked players.

This demonstrates that while an expensive army might raise the competitive level, it does not determine entry into the elite. The gap between the champions and the bottom players is defined principally by skill and strategy, not by the budget.

Meta analysis

8.1 Is there a statistically significant correlation between an army's popularity and its competitive performance?

The second area of inspection is the Meta Analysis. Specifically, the primary research question is: 'Is there a statistically significant correlation between an army's

popularity (player count) and its competitive performance (win rate and rank)?' This analysis aims to verify whether the most played factions (Mainstream) are inherently stronger than the less popular ones or if this is independent of popularity.

To test this hypothesis, we analyzed the 27 unique factions present in the tournament against two key indicators of performance: Win Rate and Mean Rank. The results obtained are:

Popularity vs Win Rate	**Result**
Pearson Correlation (r)	0.0007
Pearson p-value	0.9974
Spearman Correlation (ρ)	-0.0538
Spearman p-value	0.7898

Table 7: Correlation Analysis: Faction Popularity vs win rate

Popularity vs Mean Rank	**Result**
Pearson Correlation (r)	0.0138
Pearson p-value	0.9454
Spearman Correlation (ρ)	0.1666
Spearman p-value	0.4062

Table 8: Correlation Analysis: Faction Popularity vs mean rank

The results show a correlation of almost zero. In all cases, the correlation coefficients are insignificant, and the p-values are extremely high (indicating a non-statistically significant result). This scientifically proves that popularity is not an indicator of strength.

- The frequency of victories is independent of the number of people playing with that faction.
- The final position in the tournament (ranking) is not influenced by the popularity of the army.

Moreover, to dig deeper, the armies have been categorized into three groups based on player count;

- Mainstream (top 33%)
- Mid
- Niche (bottom 33%)

However, the mainstream performance consists of the seven most popular armies, which achieve an average win rate of 48.84% and an average rank of 205.96. On the other hand, niche performance consists of the 12 least played armies, which achieve a higher average win rate of 50.73% and a better average rank of 193.03. A T-Test comparing the win rates of mainstream vs niche groups yielded a p-value of 0.6145, which is not statistically significant. The results obtained suggest that there is no "penalty" for playing a popular army or any "bonus" for playing a rare one. Definitely, something to consider is that high popularity often dilutes the performance metrics because a faction is played by a mix of experts and beginners. In contrast, niche armies are usually played by specialists. In conclusion, the analysis confirms that the tournament meta is diverse and viable. Players do not need to follow it to win.

8.2 Does the choice of Detachment significantly influence the Win Rate?

In Warhammer 40k, choosing an army is only the first step, players must also select a "Detachment" (a specific set of rules and abilities) that defines their playstyle. For this reason, a natural question is: "Does the choice of Detachment significantly influence the Win Rate?"

To ensure the statistical robustness of this analysis, some strict filters have been applied to the dataset. First of all, the analysis has focused on factions with at least 15 players, and within those, only detachments fielded by a minimum of 5 players have been analyzed. This approach filters out part of the noise and isolates outliers, ensuring that the results reflect genuine meta trends.

The three most significant case studies are presented in the proposed study:

Detachment	**Win Rate**	**95% CI**
Skitarii Hunter Cohort	61.25%	±10.62%
Haloscreed Battle Clade	56.47%	±7.53%

Table 9: Detachment Performance: Adeptus Mechanicus (p-value: 0.4674)

Detachment	**Win Rate**	**95% CI**
Spearhead-at-arms	48.44%	±14.96%
Questoris Companions	35.42%	±17.23%
Valourstrike Lance	31.25%	±5.48%

Table 10: Detachment Performance: Imperial Knights (Best vs Worst p-value: 0.0869)

Detachment	**Win Rate**	**95% CI**
Aspect Host	57.24%	±7.09%
Warhost	47.22%	±15.70%

Table 11: Detachment Performance: Aeldari (p-value: 0.1972)

The analysis reveals that the choice of the Detachment is often as critical as the choice of the faction.

- **Adeptus mechanicus**: Considering the three cases analyzed, the Skitarii Hunter Cohort represents the peak of the tournament meta. The win rate is about 61.25% (±10.6% CI).
- **Imperial knights**: The Spearhead-at-arms detachment achieves a win rate of 48.44%, while the other two detachments have lower performance. Specifically, Questories Companions 35.42% and Valourstrke Lance 31.25%. The effect size, measured using Cohen's d > 1.0, confirms that this is not a minor variation, and the performance gap is extreme.

- **Aeldari**: Aspect Host achieves a win rate of 57.24% with a tight confidence interval of ±7.1%. This indicates highly reliable performance. On the other hand, Warhost has a lower win rate, around 47.22%, but has a high confidence interval of ±15.70%. This disparity reveals that while the Aspect Host is a stable competitive choice, the Warhost is more volatile, making it a riskier choice.

The Detachment analysis reveals the hidden layer of the meta. While general faction win rates suggest balance, the internal data exposes deep disparities. Despite the limitations of the dataset size (resulting in higher p-values), the effect size analysis validates the existence of a better build.

8.3 Summary of meta analysis

In conclusion, the Meta Analysis makes a distinct contrast between the macro level and micro level of the tournament.

- **Macro level (Factions)**: The first analysis proves that popularity is not a predictor of performance.
- **Micro level (Detachments)**: The internal analysis reveals that the specific choice of detachment acts as a significant multiplier of success. Despite statistical limitations due to sample size, the effect sizes confirm that choosing the correct Detachments provides a structural advantage, offering higher consistency and win potential.

Unit analysis

9.1 Do indispensable units exist?

After analyzing factions and detachments, the third analysis concerns individual units. A common belief in the Warhammer community is the existence of core units, essential models that every competitive list must contain in order to be valid.

Unit	Pop %	WR With	WR Without
Fire Dragons	96.8%	53.27%	50.00%
Howling Banshees	93.5%	54.13%	39.29%
Warp Spiders	93.5%	54.13%	39.29%
Fuegan	90.3%	54.02%	45.24%
Rangers	90.3%	54.72%	38.69%
Wave Serpent	90.3%	54.02%	45.24%
Jain Zar	87.1%	54.63%	43.30%
Lhykhis	87.1%	53.70%	49.55%
Swooping Hawks	83.9%	52.88%	54.64%
Corsair Voidreavers	61.3%	53.01%	53.42%
Dark Reapers	51.6%	51.23%	55.24%
Falcon	51.6%	48.44%	58.21%

Table 12: Unit Impact Analysis: Aeldari

Unit	Pop %	WR With	WR Without
Jakhals	100.0%	50.00%	–
Kharn the Betrayer	100.0%	50.00%	–
Khorne Berzerkers	100.0%	50.00%	–
Slaughterbound	91.3%	50.00%	50.00%
Chaos Spawn	87.0%	48.12%	62.50%
Exalted Eightbound	87.0%	49.38%	54.17%
Chaos Rhino	82.6%	49.34%	53.12%
Eightbound	82.6%	49.34%	53.12%
Daemon Prince (Khorne)	73.9%	50.00%	50.00%
Forgefiend	65.2%	50.00%	50.00%

Table 13: Unit Impact Analysis: World Eaters

Unit	Pop %	WR With	WR Without
Intercessor Squad	87.5%	51.91%	40.62%
Lieutenant (Combi)	78.1%	49.64%	53.57%
Assault Intercessors (JP)	68.8%	48.46%	55.00%
Inceptor Squad	59.4%	48.21%	53.85%
Centurion Devastators	56.2%	50.89%	50.00%
Vanguard Veterans (JP)	53.1%	49.47%	51.67%

Table 14: Unit Impact Analysis: Space Marines

Firstly, the top 5 most played armies have been analyzed, and the core units have been defined as units included in at least 50% of lists. The results suggest that some factions, such as the Aeldari and the World Eaters, contain many core units. For example, 100% of World Eaters players use two units (*Khorne Berzerkers* and *Jakhals*). However, other factions, such as Space Marines, have few core units.

To test whether the popular units are actually indispensable, we performed a local test within each of the top 5 factions, comparing the performance of players with core units to those without them.

The results obtained indicate that none of the five factions analyzed achieved statistically significant results ($p > 0.05$). It is essential to keep in mind that this specific analysis was severely limited by the sample size, both in terms of core units and niche units. The most commonly used units (used by at least 50% of players) are utilized by the entire spectrum of participants, ranging from podium to the lowest ranked players. Consequently, their win rates will naturally converge toward the faction average, making it difficult to isolate their specific contribution to victory. Conversely, niche units are subject to high variance due to the limited number of observations. For example, if a rare unit is selected by only a small group of highly skilled players, it may appear statistically superior (as a false positive) due to the players skills.

Therefore, the lack of statistical significance suggests that, given this sample size, it cannot mathematically prove that core units provide a win rate advantage. To isolate unit strength from player skill, a significantly larger dataset is required. This would allow for a sufficient number of observations for niche units to stabilize their variance, enabling a fair comparison against the core choices. However, it is worth noting the presence of specific units with a frequency between 90% and 100% among players of the same faction, which effectively signals the existence of "must have" troops in terms of army construction.

9.2 Do elite armies with fewer high-point units outperform swarm armies with many low-point units?

The final question of the unit analysis investigates the strategic archetype of the army list. The research question is "Do elite armies with fewer high-point units outperform swarm armies with many low-point units?". To define the strategic archetype of each army, we considered two fundamental variables for each player:

- Average points per unit: used to measure the average points of the units selected by the player
- Total model count: used to measure the total number of individual miniatures (models) fielded in the army, representing the "board presence".

Combining these metrics, we obtain the quality ratio ($\frac{\text{Average Points}}{\text{Total Models}}$). This ratio is used as a metric to classify army composition. It helps to determine whether certain factions tend to use a large number of units with low points (horde) or a concentrated force of a few units with high points (elite).

Army	Strategy	N	WR (%)
Space Marines	QUALITY	16	50.00%
	QUANTITY	16	51.00%
Aeldari	QUALITY	15	60.83%
	QUANTITY	16	45.98%
Adeptus Mechanicus	QUALITY	14	58.29%
	QUANTITY	14	58.93%
World Eaters	QUALITY	10	51.25%
	QUANTITY	13	49.04%
Death Guard	QUALITY	11	42.05%
	QUANTITY	12	42.11%

Table 15: Strategy Comparison: Quality vs. Quantity Lists

The study analyzed the five most popular factions and obtained the following results:

- For 4 out the 5 analyzed factions have no statistical impact on performance. These results indicate high strategic versatility. These factions are well designed to support multiple playstyles.
- The **Aeldari** faction stands out as the only statistically significant one. In this faction, the quality build WR is 60.83% while the quantity build WR is approximately 45.98%. The difference is +14.85% ($p = 0.0233$) with a large effect size. Unlike the other factions analyzed, the Aeldari appear to perform worst when played as a horde. The data prove that the optimal way to play the Aeldari is the elite archetype.

Global Strategy	N	Win Rate (%)
QUALITY	201	48.03%
QUANTITY	201	50.66%

Table 16: Global Strategy Comparison: Elite vs. Horde Lists

After analyzing specific factions, a global analysis has been performed across all 402 players to determine whether the game tends to prefer one playstyle. The global analysis yielded a p-value of 0.1440, which is statistically non significant, and an effect size of -0.146, this means it is negligible.

On a macro level, the tournament is balanced. The win rate for "quality" players (48.03%) is almost identical to that of "quantity" players (50.66%). This confirms that Warhammer 40k does not tend to favor elite armies over horde or vice versa). The advantage depends entirely on the specific mechanics of the chosen faction.

9.3 Summary of unit analysis

The analysis reveals that, although the composition of the army determines the competitive basis, it does not guarantee victory.

- The first analysis revealed the presence of certain core units, units used by over 90% of players, which define the use of certain standard units. However, their presence shows no statistical correlation with higher win rates, implying that player skill remains the decisive factor once units have been selected.
- Overall, the game does not favor either Elite or Horde strategies ($p=0.14$), indicating a healthy structural balance. However, the exception of the Aeldari demonstrates that there are some faction specific limitations that can lead to a competitive disadvantage.

Faction dynamic

10.1 Does the game have a similar dynamic to the game Rock, Paper, Scissors?

After analyzing the economic impact, the general army performance, and the efficiency of individual units, this fourth and final section investigates the direct interactions between factions. The research question addresses the core of the game's balance: Does a "Rock-Paper-Scissors" dynamic exist? Specifically, are there statistically proven hard counters where one faction holds an advantage over another?

To answer this question, a matchup matrix has been generated involving the Top 10 armies, analyzing a total of 40 unique pairwise combinations.

Faction	Best Matchup	WR (%)
Space Marines	Imperial Knights	85.7*
Eldari	Death Guard	80.0
Ad. Mechanicus	T'au Empire	87.5
World Eaters	Necrons	83.3
Death Guard	Imperial Knights	66.7
Imperial Knights	Necrons	62.5
Necrons	Chaos S. Marines	100.0*
Chaos S. Marines	Eldari	71.4
T'au Empire	World Eaters	66.7
Adepta Sororitas	Death Guard	80.0

Table 17: Faction Analysis: Most Favorable Matchups (* = Significant)

Faction	Worst Matchup	WR (%)
Space Marines	Ad. Mechanicus	38.9
Eldari	Chaos S. Marines	28.6
Ad. Mechanicus	Chaos S. Marines	54.5
World Eaters	Ad. Mechanicus	31.2
Death Guard	Necrons	0.0*
Imperial Knights	Space Marines	14.3*
Necrons	World Eaters	16.7
Chaos S. Marines	Necrons	0.0*
T'au Empire	Ad. Mechanicus	12.5
Adepta Sororitas	Eldari	33.3

Table 18: Faction Analysis: Most Unfavorable Matchups (* = Significant)

The analysis of the pairings revealed a complex situation, with a distinction between statistical certainty and observed trends. Only 3 out of 40 matchups (7.5%) reached the threshold for statistical significance ($p < 0.05$). This low number is heavily influenced by the sample size fragmentation. While the total dataset includes 1584 matches, dividing them across specific faction pairings reduces the observations for individual cells, making it harder to achieve statistically significant results. However, the absence of statistical significance should not be mistaken for perfect balance. The matrix reveals several directional trends where there are matchups with >60% or <40% win rates that, while not meeting the $p < 0.05$ criteria, suggest underlying advantages. In conclusion, while there are only three cases in which we can consider hard counters, the meta also contains some soft counters. These matchups represent distinct advantages that influence the game, even if the current sample size prevents a definitive statistical verdict.

The analysis now examines the specific results that achieved statistical significance:

- **Necrons**: The necrons faction exhibits the most polarized performance, achieving a 100% win rate against both chaos space marines and death guard. This suggests that these specific armies struggle against the necrons.
  A reasonable explanation for these results is that Necrons are an extremely specialized long-range shooting faction, while both Chaos Space Marines and Death Guard are primarily melee-focused armies with very limited artillery units, resulting in the fact that the two factions are unable to close the distance, with many units being killed beforehand.
- **Space marines**: The space marines faction demonstrates a dominant 85.7% win rate against the imperial knights.
  Space Marines and Imperial Knights are two highly distinctive factions: the former represent the most versatile army in the game, with a wide variety of unit types, while the latter consist exclusively of very expensive vehicles specialized almost entirely in direct combat. This matchup highlights how, in Warhammer 40K, raw battlefield dominance is not the sole factor in securing victory. Instead, success is often determined by the ability to score victory points through controlling key objectives and completing secondary missions, an area where a more flexible and adaptable faction tends to excel.

Nevertheless, the tournament that has been analyzed also presents numerous fairly balanced matches:

- **Aeldari Consistency**: The Aeldari faction emerges as a benchmark for competitive equilibrium. The faction against Space Marines, which is the tournament's most frequent pairing, resulted a 50% win rate. This trend of exact balance extended to matchups against both the T'au Empire and Imperial Knights, suggesting that games involving Aeldari are decided by player skill rather than faction advantage.

10.2 Summary of faction dynamic

In conclusion, the analysis confirms the presence of specific hard counters, proving that matchup advantages exist. However, based on the current data, the meta does not appear to be defined by a rigid "Rock-Paper-Scissors" chain.

Moreover, the matrix reveals several directional trends that strongly suggest other imbalances exist. However, the small volume of these trends does not tend to be statistically significant. Therefore, gathering more observations would be important to validate these patterns.

Limitations and future developments

The conducted analysis presents several methodological and contextual limitations that are important to acknowledge for a correct interpretation of the results:

Sampling limited to a single tournament The primary limitation concerns the choice to analyze exclusively the Warhammer 40,000 World Championship 2025 (Atlanta, GA). Although this is a prestigious event with 402 participants, a significant sample for a single tournament, this choice inevitably limits the

generalizability of the results.

The decision to focus on a single tournament was driven by a fundamental technical challenge of the game. Games Workshop frequently updates unit point values through periodic balance patches, making it extremely difficult to compare tournaments played at different times. Expanding the dataset to include large tournaments with different point configurations would have required complex adjustments and normalizations that could have introduced systematic biases and compromised the accuracy of the economic analysis.
This limitation reduces statistical robustness in some specific areas:

- Faction matchup analysis: Only 3 of the 40 analyzed pairings (7.5%) reached statistical significance ($p < 0.05$), primarily due to sample fragmentation when divided by specific faction pairs
- Detachment analysis: Despite relevant effect sizes (Cohen's $d > 1.0$ in some cases), several comparisons present p-values above the conventional significance threshold
- Geographic and temporal variability: The results reflect the specific meta-game of November 2025 and may not represent valid dynamics in other geographic contexts or time periods

Army cost estimates The economic analysis is based on estimates rather than actual costs incurred by players. The implemented price-scaling system, while founded on reasonable logic (tournament points/base points ratio, thresholds for enhancements), introduces a margin of approximation that may not perfectly reflect players' actual purchases. Furthermore, some participants may have used miniatures already owned, purchased at discounted prices in bundle boxes, or obtained in the second-hand market.

Data cleaning dictionary limitations The curated mapping dictionary used to standardize unit names and resolve naming inconsistencies related to player submissions was developed specifically for this tournament's dataset. While it successfully achieved 100% matching rate for the World Championship data, it may be insufficient for cleaning datasets from other tournaments.

Despite these limitations, the analysis still provides valuable insights into the relationship between economic investment and competitive performance, highlighting thatat least in the context of the 2025 World Championshipstrategic skill remains the dominant factor over budget spent.

The main future development of this project is to expand the dataset by extracting data from additional tournaments, while also developing methods to properly handle balance changes over time and the release of new units.
Another interesting development for the community would be the creation of an interface that allows users to build their own army by selecting faction, detachment, and units, and then evaluates its potential competitive performance by identifying, through a maximum-similarity matching system, the most comparable army or armies.

Conclusion

This study set out to investigate a fundamental question in competitive tabletop wargaming: does money buy victory in Warhammer 40,000?
By analyzing 402 participants at the 2025 World Championship, we examined the intersection of economic investment, faction choices, army composition, and competitive performance.

Our analysis reveals that while a weak positive correlation exists between army cost and performance ($r = 0.131$, $p < 0.01$), **financial investment is not the defining factor for competitive success**. The comparison between top-tier and bottom-tier players demonstrates that reaching the elite level requires no additional budgetthe top 25 players spent statistically the same amount as the bottom 25 ($p = 0.3082$). This finding decisively refutes a pure "pay-to-win" model.

The meta-game analysis confirms a **healthy competitive diversity**. Faction popularity shows no correlation with win rate ($r = 0.0007$), proving that players are not forced to follow mainstream choices to succeed. However, the analysis of detachments reveals a hidden layer of strategic depth: within individual factions, specific detachment choices can provide substantial advantages, with effect sizes exceeding Cohen's $d > 1.0$ in several cases.

The faction matchup matrix identified **limited evidence of hard counters** as only 3 of 40 pairings achieved statistical significance, suggesting that while matchup dynamics exist, they do not define a rigid rock-paper-scissors meta-game. The tournament remains fundamentally skill-based rather than determined by faction selection alone.

Ultimately, while economic factors influence accessibility (like the average investment of 927€for a competitive army) and provide marginal advantages, they do not determine outcomes at the highest levels of play. The real barrier to competitive success is not financialit is the hundreds of hours required to master army synergies, matchup dynamics, and tournament-level decision-making. In this sense, Warhammer 40,000 preserves its integrity as a competitive endeavor where intellect and preparation triumph over mere investment.

Author Contributions

This project was a collaborative effort with distinct responsibilities assigned to each team member:

- **Mezzanzanica Chiara:** Data acquisition and parsing pipeline. Developed the API extraction scripts, parsing logic, and validation framework.
- **Cesaratto Ilaria:** Data cleaning, integration, and storage infrastructure. Responsible for unit name standardization, price assignment logic, and database implementation.
- **Porro Andrea:** Statistical analysis and interpretation. Conducted economic analysis, meta analysis, unit analysis, and faction dynamic investigations.

A Appendix

Database Schema

Table	Attribute	Type	Key / Constraint
tournaments	ID_TOURNAMENT	INTEGER	Primary Key
	NAME	TEXT	Not Null
	LOCATION	TEXT	–
	EVENT_DATE	DATE	–
	TOTAL_PLAYERS	INTEGER	–
	YEAR	INTEGER	–
players	ID_PLAYER	TEXT	Primary Key
	ID_TOURNAMENT	INTEGER	Foreign Key → tournaments(ID_TOURNAMENT)
	RANK	INTEGER	–
	NAME	TEXT	–
	SURNAME	TEXT	–
	FULL_NAME	TEXT	–
	ARMY	TEXT	–
	DETACHMENT	TEXT	–
	WINS	INTEGER	–
	LOSSES	INTEGER	–
	TIES	INTEGER	–
	SCORE	TEXT	–
army_lists	ID_TOURNAMENT	INTEGER	Foreign Key → tournaments(ID_TOURNAMENT)
	ID_PLAYER	TEXT	Foreign Key → players(ID_PLAYER)
	UNITS	TEXT	Serialized structure
	WARLORD	TEXT	–
army_list_units	ID_PLAYER	TEXT	Foreign Key → players(ID_PLAYER)
	CLEAN_NAME	TEXT	Indexed
	POINTS	INTEGER	–
	QUANTITY	INTEGER	–
	SINGLE_PRICE	REAL	–
	TOTAL_PRICE	REAL	–
matches	ID_MATCH	INTEGER	Primary Key (autoincrement)
	ID_TOURNAMENT	INTEGER	Foreign Key → tournaments(ID_TOURNAMENT)
	ID_PLAYER1	TEXT	Foreign Key → players(ID_PLAYER)
	NAME_PLAYER1	TEXT	–
	ID_PLAYER2	TEXT	Foreign Key → players(ID_PLAYER)
	NAME_PLAYER2	TEXT	–
	SCORE_PLAYER1	INTEGER	–
	SCORE_PLAYER2	INTEGER	–
prices	CLEAN_NAME	TEXT	Composite PK, Indexed
	ARMY	TEXT	Composite PK
	POINTS_UNIT	INTEGER	–
	PRICE_UNIT	REAL	–

Table 19: Relational schema of the final SQLite database. Primary keys uniquely identify entities, while foreign keys enforce referential integrity across tournaments, players, matches, and army compositions. The prices table serves as a reference dataset for unit cost estimation.

Operational guide - Pipeline Architecture

The pipeline consists of 6 main scripts that must be executed sequentially and two ipynb that come after:

Data Acquisition

- **API_acquisition.py** – Extracts raw tournament data from ListEngine API.
  Input:
  1. event_id (e.g., abc123xyz) – Event identifier of the tournament of interest taken from the URL on https://listengine.onrender.com/events

  Output:
  1. warhammer_extended_lists.json – Raw tournament data
- **parser.py** – Initial parsing of raw JSON data into structured format.
  Input:
  1. warhammer_extended_lists.json

  Output:
  1. warhammer_valid_lists.json – Successfully parsed army lists
  2. warhammer_invalid_for_reparse.json – Lists requiring secondary parsing
- **reparser.py** – Secondary parsing pass to correct extraction errors.
  Input:
  1. warhammer_extended_lists.json
  2. warhammer_invalid_for_reparse.json

  Output:
  1. warhammer_reparsed_valid.json – Successfully reparsed lists
  2. warhammer_still_invalid.json – Lists still requiring manual intervention
- **merger.py** – Merges parsed data with manual corrections.
  Input:
  1. warhammer_valid_lists.json – Initial parsing results
  2. warhammer_reparsed_valid.json – Secondary parsing results
  3. warhammer_manual_lists.json – Manually compiled lists (19 lists)

  Output:
  1. warhammer_merged_lists.json – Combined dataset
  2. warhammer_final_valid.json – Validation of the combined dataset
  3. warhammer_final_invalid.json – Empty file as all lists are valid
- **cleaner.py** – Final data cleaning and standardization.
  Input:
  1. warhammer_final_valid.json

  Output:
  1. warhammer_final_cleaned.json – Final clean dataset ready for analysis
- **validator.py** – Functions used throughout the pipeline to locate invalid lists
  (This file is imported by other scripts and is not executed standalone)

Database Creation and Loading

After the data cleaning phase, the final dataset is transformed into a normalized relational database through an additional script.

- **Warhammer_SQL.py** – Creation and loading of the SQLite database. Its purpose is to integrate the cleaned JSON tournament dataset with the external unit price CSV file and store the resulting data in a structured relational database, ensuring data integrity, normalization, and reproducibility.

  Input:

  1. warhammer_final_cleaned.json – Final cleaned tournament dataset
  2. Warhammer_prezzi.csv – Unit price reference table

  Main Operations:

  - Creation of the SQLite database schema with primary and foreign key constraints
  - Normalization of tournament, player, match, and army list data
  - Explosion of army lists into unit-level observations
  - Deterministic price estimation based on unit points
  - Automatic index creation for query optimization

  Output:

  1. warhammer_tournaments.db – SQLite database containing structured tournament data

Data Quality Analysis

After the database creation phase, a dedicated data quality assessment is performed to characterize the quality of the unit price data integrated into the database, focusing on completeness, consistency, and reliability of the cost information.

- **data_quality_assessment.ipynb** – Data quality analysis.

  Input:

  1. Warhammer_prezzi.csv – Raw unit price reference table
  2. warhammer_final_cleaned.json – Cleaned tournament data before integration
  3. warhammer_tournaments.db – SQLite database containing integrated data

  Main Operations:

  - Evaluation of missing values and invalid entries in pricing data
  - Consistency checks on unit names and point values across sources
  - Identification of duplicates and anomalous price observations

  Output:

  1. Data quality metrics and summary tables

Data Analysis

- **warhammer_analysis.ipynb** – Exploratory and analytical evaluation of the relationship between army cost and competitive performance.

  Input:

  1. warhammer_tournaments.db – SQLite database with integrated tournament data

  Output:

  1. Descriptive statistics, visualizations, and analytical results.